cultivating, we procured almost everything, but ‘O’ rings, the specialised rubber joints made by a company called Le Joint Francais, remained elusive.

With his incredible contacts in France, Kanwar, who is better known in India for his vineyards than his invaluable contribution to India’s space programme, managed to get them. It still remains a mystery how he collected the few ‘O’ rings from the French company; Kanwal told me that he casually carried them in his suitcase during one of his many monthly flights from France to India.

By 1981, my team put together one-and-a-half Vikas engines—a full engine that was ready to be tested and, with whatever leftover material we had, we put together a turbo pump and the upper part of the engine. This, however, did not qualify us to get into PSLV, since the Vikas engine—though we had made one—was not officially a project yet. Anything that has to go into a rocket has to be approved as a project, and the product has to be tested before being incorporated into a rocket.

The one-and-a-half Vikas engine we made was out of Dhawan’s discretionary funds. Without a project status, it could not be tested. Even if it is given the clearance, there was no facility in India to ground-test the engine. We had to break it down, take the parts to Vernon, France and re-assemble before testing. That was to happen more than six years later, in 1985, when India’s liquid propulsion capabilities came of age.

But now the Gowarikar-Muthunayagam tussle has come to such a stage that Dhawan had to create a separate unit—the Auxiliary Propulsion Systems Unit (APSU)—to accommodate Muthunayagam, who would now report directly to the ISRO chairman. APSU would transform to become the Liquid Propulsion Systems Unit, and later what is today called the Liquid Propulsion Systems Centre.

APSU worked on small control rockets and flow components. Major liquid propulsion systems came directly under VSSC. This put me in a spot because of my sphere of work, on Vikas and other liquid propulsion systems came under VSSC. When I pointed this out to Muthungayagam, he told me to report to Gowarikar. I did not mind. I thought Gowarikar would